

GALAPAGOS ISLANDS

The island group that inspired the theory of evolution

The Galapagos is a remote archipelago of volcanic islands in the vast blue of the Pacific Ocean. These islands formed when the Earth's crust moved over a hot spot in the mantle (a warmer layer of semisolid rock below the crust), causing it to melt and form a sequence of volcanoes. Many erupting volcanoes reached the sea's surface before becoming extinct, and the cooled lava created the Galapagos Islands. Due to their remoteness from other land masses, these islands have unique endemic species found nowhere else on Earth.

The story of evolution

Charles Darwin, a British naturalist, traveled to the Galapagos Islands in 1835. What he saw there helped him develop his theory of evolution by natural selection, whereby species change as generations pass. Darwin saw fascinating examples of divergence—where animals

on different islands had developed sufficient differences to become separate species. As a result, the Galapagos Islands have many unique animals, such as the various finches and giant tortoises.

Three major currents converge here, bringing nutrients and plankton that support fish, marine mammals, and seabirds. The islands are periodically affected by El Niño, which causes warmer and wetter conditions that can benefit plants and land animals, but is devastating to marine life.



PIONEER SPECIES

The lava cactus is quick to colonize the bare rock of lava fields. Its fleshy stems store water and are a valuable source of moisture for wildlife including land iguanas and tortoises. Cactus finches assist in populating new areas by distributing its seeds.

FLIGHTLESS CORMORANT

DIFFICULT TO ESCAPE
Historically, there were no land-based predators on the Galapagos Islands and so this cormorant lost the ability to fly. Since the arrival of humans, its numbers have decreased because of introduced predators such as cats, dogs, and pigs.

Approximately 127 islands and islets in total

LAVA CACTUS

SALLY LIGHTFOOT CRAB
A World Heritage Site since 1978

CLEANER CRABS

Bright red Sally Lightfoot crabs have a mutually beneficial relationship with marine iguanas. They keep the resting iguanas clean by picking ticks and algae off their skin. The crabs eat what they find and so receive a reward for their work.